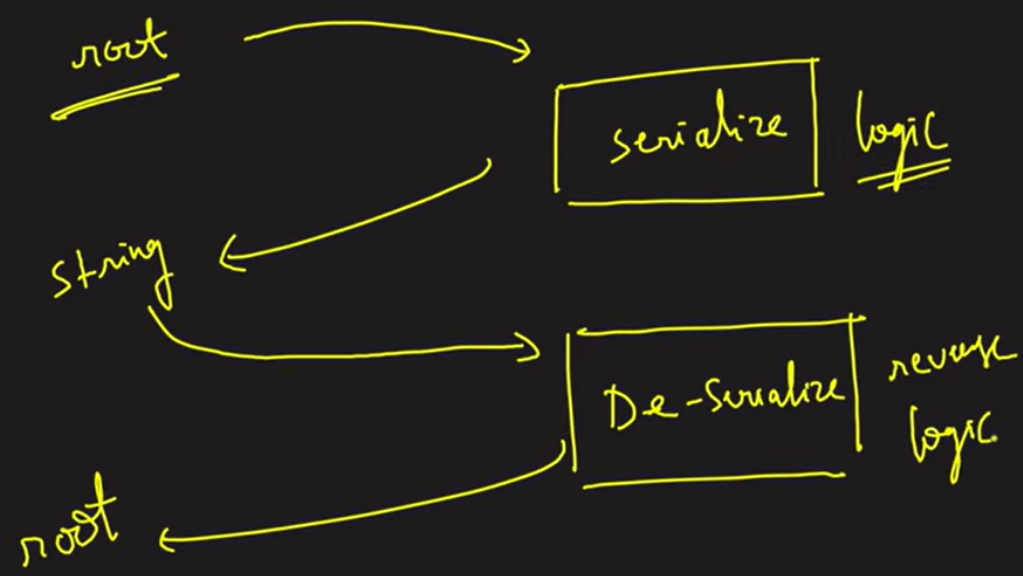
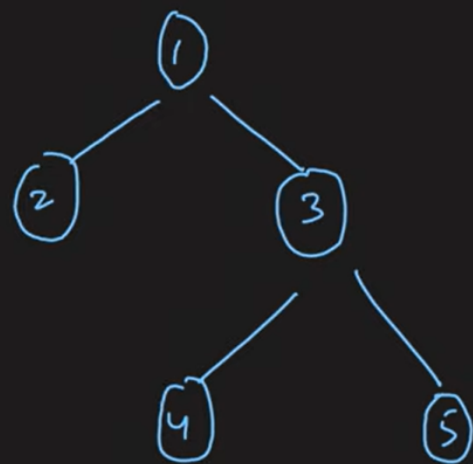
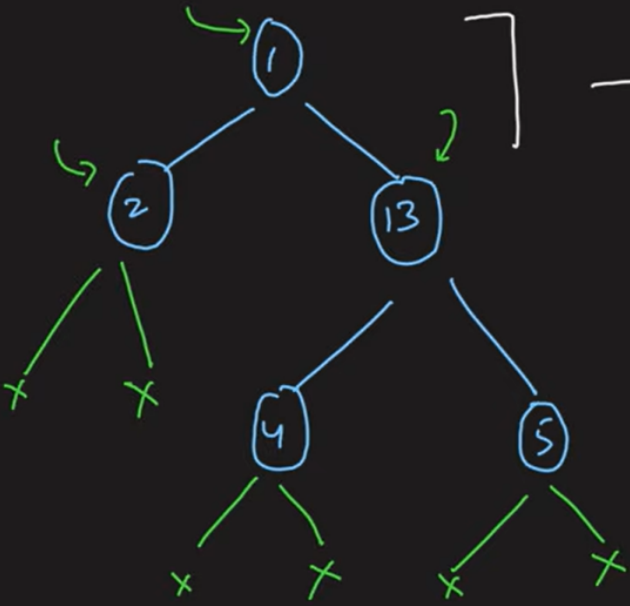


Serialize & De-Serialize Binary Tree



Serialize & De-Serialize Binary Tree



Serialize → LO

String → 1, 2, 13, #, #, 4, 5, #, #, #, #,

5

4

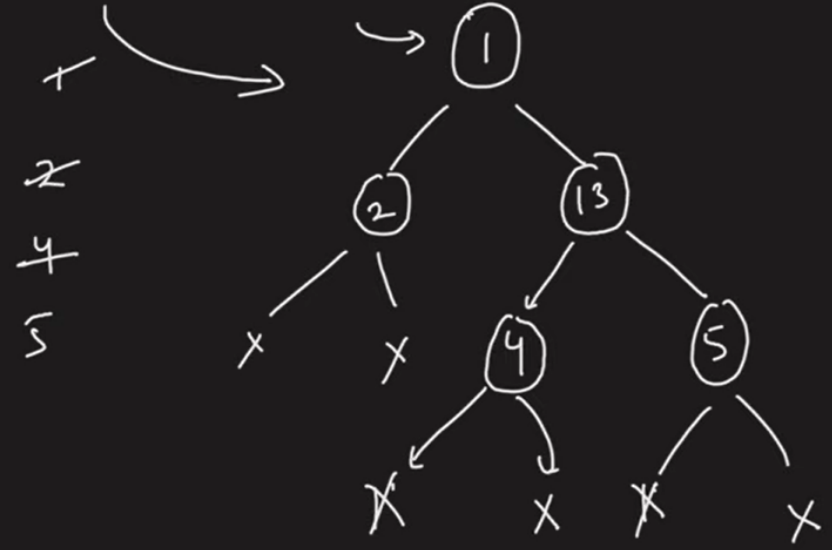
13

2

1

+

0 → empty



2.00

C++

Autocomplete

```
11 public:
12
13 // Encodes a tree to a single string.
14 string serialize(TreeNode* root) {
15     if(!root) return "";
16
17     string s = "";
18     queue<TreeNode*> q;
19     q.push(root);
20     while(!q.empty()) {
21         TreeNode* curNode = q.front();
22         q.pop();
23         if(curNode==NULL) s.append("#,");
24         else s.append(to_string(curNode->val)+' ');
25         if(curNode != NULL){
26             q.push(curNode->left);
27             q.push(curNode->right);
28         }
29     }
30     cout << s;
31     return s;
32 }
33
34 // Decodes your encoded data to tree.
35 TreeNode* deserialize(string data) {
36     if(data.size() == 0) return NULL;
37     stringstream s(data);
38     string str;
39     getline(s, str, ',');
40     TreeNode *root = new TreeNode(stoi(str));
41     queue<TreeNode*> q;
42     q.push(root);
43     while(!q.empty()) {
44
45         TreeNode *node = q.front();
46         q.pop();
47
48         getline(s, str, ',');
49         if(str == "#") {
50             node->left = NULL;
51         }
```

```
2.00
C++
Autocomplete

33
34 // Decodes your encoded data to tree.
35 TreeNode* deserialize(string data) {
36     if(data.size() == 0) return NULL;
37     stringstream s(data);
38     string str;
39     getline(s, str, ',');
40     TreeNode *root = new TreeNode(stoi(str));
41     queue<TreeNode*> q;
42     q.push(root);
43     while(!q.empty()) {
44         TreeNode *node = q.front();
45         q.pop();
46
47         // Left child
48         getline(s, str, ',');
49         if(str == "#") {
50             node->left = NULL;
51         }
52         else {
53             TreeNode* leftNode = new TreeNode(stoi(str));
54             node->left = leftNode;
55             q.push(leftNode);
56         }
57
58         // Right child
59         getline(s, str, ',');
60         if(str == "#") {
61             node->right = NULL;
62         }
63         else {
64             TreeNode* rightNode = new TreeNode(stoi(str));
65             node->right = rightNode;
66             q.push(rightNode);
67         }
68     }
69     return root;
70 }
71
72 // Your Codec object will be instantiated and called as such:
73 // Codec ser, deser;
```

